

5. Second order differential equations		
5.1	The linear second order differential equation $a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = f(x)$ where a, b and c are real constants and the particular integral can be found by inspection or trial.	The auxiliary equation may have real distinct, equal or complex roots. $f(x)$ will have one of the forms $k e^{px}$, $A + Bx$, $p + qx + cx^2$ or $m \cos \omega x + n \sin \omega x$. Students should be familiar with the terms ‘complementary function’ and ‘particular integral’. Students should be able to solve equations of the form $\frac{d^2y}{dx^2} + 4y = \sin 2x$.
5.2	Differential equations reducible to the above types by means of a given substitution.	